AI Explainability POV of Financial Regulators

Tisha Evite

Pseudonym	Region	Role / Archetype
Juan (P21)	Asia	Developer for Innovation Projects (Central Bank)
Omar (P22)	Middle East	Analytics & AI Leader (Supervisory Authority)
Bat (P23)	Asia	Banking Supervisor (Central Bank)
Gabriel (P24)	Asia	Bank Officer for AML Compliance (Central Bank)
Timur (P25)	Central Asia	Automation Developer (Central Bank)
Nia (P26)	Africa	Data Scientist (Central Bank)
Thomas (P28)	Europe	Senior Advisor in Horizontal Supervision (Central Bank)
Oliver (P30)	Europe	Data Scientist (Conduct Regulator)
Arthur (P31)	Global / Europe	Research Fellow (International Prudential Body)

Dual lenses

Regulators expectations and experiences of AI are shaped:

1. **Internally:** as users, how they deploy AI for their own supervisory work
2. **Externally:** how they regulate financial institutions that use AI

Internal XAI (1)

➔ **Laymanization for senior policymakers**

Ultimately concerned with the results and the **big picture.** No SHAP or bee plots for surrogate models. Use simple narratives and basic bar graphs.

➔ **Economic causality**

Base on **financial or economic theory.** CBs restrict advance ML to data gathering and prefer high interpretable models.

The challenge…we said a lot, but the challenge really is to have an explainability that is laymanized in a way that anyone can understand.

- *Developer, Central Bank, Asia*

Interpretability meaning that you are able to interpret, Shapley values as a physical coefficient, like close to causality… according to economic laws, economic principles, and national principles, so not just statistical conclusions, like Shapley values, statistical artifacts.

- *Developer, Central Bank, Europe*

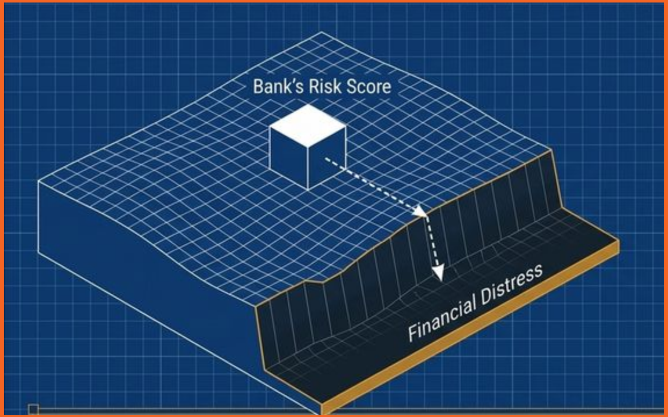
Internal XAI (2)

➔ **Source traceability**

For Gen AI or LLM, explainability is evidencing. Supervisors expect AI to point to sources, exact test, and clickable links. **Ground the logic in applicable frameworks** to verify the decision against their own expertise.

➔ **Counterfactuals**

SHAP is "backward looking" and supervisors need "forward" too. E.g. **determine the smallest change in the predictors required for a change in result**...for a bank to down/upgraded or for banks, match with internal submission-



Lexicon of developer vs regulator

Regulator / supervisor	Technical / developer	Outcome
Accountability and fairness	High F1 or accuracy	Explicit human sign-off workflows
Auditability and traceability	Feature importance (SHAP/LIME), technical documentation	Step-by-step tracing back documentation
Systemic risk prevention	Fully autonomous agents	Hardcoded circuit breake and HITL thresholds
Ground truth / no hallucination	Automated NLP Parsing	Source-citation links mapped to regulatory guidelines or memos

I had to go to technical level and the econometricians…in my opinion are also very sophisticated technically, but they have their own mindset based on linear models... It is difficult to convince them of a new approach. A new approach that is not based on fixed coefficients, test p-values that do not exist, for random fo-, for machine learning models.

Audience 1: Technical peer

We are doing Python programming around the foundational model that does that. We're setting up our own sort of databases all on cloud infrastructure... So we define every single part of our process so it fits in with our existing architecture and can be plugged in quite naturally to the workflow of existing supervisors

Audience 2: Technical devploper

So the first thing that is important for us as regulators is the accountability. So on the final decision, human decision makers must remain responsible for the decision that these models have made... because you cannot fault an AI if they misreported or have maliciously reported people... who will be accountable when that happens?

Audience 3: Non-technical

External explainability

What are our takeaways as a supervised bank?

Liability sits with us.

- Both AI vendors and regulators emphasize risk and legal liability of an AI algorithm sitting with the financial institution.
- Re: “right to explanation”
  - If a bank uses AI for automated decision-making (like loan pricing or rejections), it must be prepared to provide clear explanations directly to consumers who challenge the outcome or request the rationale behind the decision
  - Again, level of detail depends on audience: regulator vs customer

They won’t explicitly tell what or what not to use.

- Laws are deliberately vague to be blanket/generally applicable as tools improve
- Stance of most conduct authorities are “technology agnostic”
- Focus on <u>proving we do not harm consumers</u>

The only thing they have to do is demonstrate to us at the end of the day that it's not systemically or systematically harming individuals

Results over back-end sophistication

- Time-sensitive operations, speed and accuracy matter over technical intricacies
  - For fraud, bank should be able to act on it real-time
- Narrative-focus on suspicious transactions
  - Articulate who, what, when, why of a transaction
  - These expectations exist regardless of whether machine or human flags it

Next steps

Credit memo pilot survey on XAI

Systematic lit review with Saban

June - August
June 4 - Aug 8 please contact me via patriciamarcella.evite@unina.it

GAIA pilot survey

After "summer"

Technical report on trustworthy XAI

December 2026

Thank you.